ELEVEN PROPERTIES OF A MORPHISM OF ALGEBRAIC SPACES ARE INSENSITIVE TO A FINITE ORDER THICKENING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2043-2064 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2066-2146, the complete author proof (79 body lines).

Editorial note. Navigation stubs: 12 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale charts for algebraic spaces, sheaves of differentials, and the schemes version of the thickening lemma; the chart reduction and the remaining cases are retained.

Source reference. Original lemma lemma-differential-product: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1227–1240. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-differentials-relative-immersion-section: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1180–1191. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-thicken-property-morphisms: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2001–2015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-thickening-scheme: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1730–1733. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-thicken-property-morphisms-cartesian: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2146–2146. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma morphisms-lemma-differentials-diagonal: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2146–2146. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-finite-integral: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2146–2146. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-lemma-base-change-immersions: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2146–2146. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-base-change-integral: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2146–2146. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-base-change-monomorphism: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2146–2146. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-base-change-proper: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2146–2146. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-remark-immersion: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2146–2146. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $(f, f') : (X \subset X') \to (Y \subset Y')$ be a morphism of thickenings of algebraic spaces over S such that $X = Y \times_{Y'} X'$. If $X \subset X'$ is a finite order thickening, then*

(1) *f is a closed immersion if and only if f' is a closed immersion,*
(2) *f is locally of finite type if and only if f' is locally of finite type,*
(3) *f is locally quasi-finite if and only if f' is locally quasi-finite,*
(4) *f is locally of finite type of relative dimension d if and only if f' is locally of finite type of relative dimension d,*
(5) *$\Omega_{X/Y} = 0$ if and only if $\Omega_{X'/Y'} = 0$,*
(6) *f is unramified if and only if f' is unramified,*
(7) *f is proper if and only if f' is proper,*
(8) *f is a finite morphism if and only if f' is an finite morphism,*
(9) *f is a monomorphism if and only if f' is a monomorphism,*
(10) *f is an immersion if and only if f' is an immersion, and*
(11) *add more here.*

Proof. Choose a scheme V' and a surjective étale morphism $V' \to Y'$. Choose a scheme U' and a surjective étale morphism $U' \to X' \times_{Y'} V'$. Set $V = Y \times_{Y'} V'$ and $U = X \times_{X'} U'$. Then for étale local properties of morphisms we can reduce

to the morphism of thickenings of schemes $(U \subset U') \to (V \subset V')$ and apply More on Morphisms, Lemma more-morphisms-lemma-thicken-property-morphisms-cartesian. This proves (2), (3), (4), (5), and (6).

The properties of morphisms in (1), (7), (8), (9), (10) are stable under base change, hence if f' has property $\mathcal{P}$, then so does f. See Spaces, Lemma spaces-lemma-base-change-immersions, and Morphisms of Spaces, Lemmas spaces-morphisms-lemma-base-change-proper, spaces-morphisms-lemma-base-change-integral, and spaces-morphisms-lemma-base-change-monomorphism.

The interesting direction in (1), (7), (8), (9), (10) is to assume that f has the property and deduce that f' has it too. By induction on the order of the thickening we may assume that $Y \subset Y'$ is a first order thickening, see discussion on finite order thickenings above.

Proof of (1). Choose a scheme V' and a surjective étale morphism $V' \to Y'$. Set $V = Y \times_{Y'} V'$, $U' = X' \times_{Y'} V'$ and $U = X \times_Y V$. Then $U \to V$ is a closed immersion, which implies that U is a scheme, which in turn implies that U' is a scheme (Lemma lemma-thickening-scheme). Thus we can apply the lemma in the case of schemes (More on Morphisms, Lemma more-morphisms-lemma-thicken-property-morphisms-cartesian) to $(U \subset U') \to (V \subset V')$ to conclude.

Proof of (7). Follows by combining (2) with results of Lemma lemma-thicken-property-morphisms and the fact that proper equals quasi-compact + separated + locally of finite type + universally closed.

Proof of (8). Follows by combining (2) with results of Lemma lemma-thicken-property-morphisms and using the fact that finite equals integral + locally of finite type (Morphisms, Lemma morphisms-lemma-finite-integral).

Proof of (9). As f is a monomorphism we have $X = X \times_Y X$. We may apply the results proved so far to the morphism of thickenings $(X \subset X') \to (X \times_Y X \subset X' \times_{Y'} X')$. We conclude $X' \to X' \times_{Y'} X'$ is a closed immersion by (1). In fact, it is a first order thickening as the ideal defining the closed immersion $X' \to X' \times_{Y'} X'$ is contained in the pullback of the ideal $\mathcal{I} \subset \mathcal{O}_{Y'}$ cutting out Y in Y'. Indeed, $X = X \times_Y X = (X' \times_{Y'} X') \times_{Y'} Y$ is contained in X'. The conormal sheaf of the closed immersion $\Delta : X' \to X' \times_{Y'} X'$ is equal to $\Omega_{X'/Y'}$ (this is the analogue of Morphisms, Lemma morphisms-lemma-differentials-diagonal for algebraic spaces and follows either by étale localization or by combining Lemmas lemma-differentials-relative-immersion-section and lemma-differential-product; some details omitted). Thus it suffices to show that $\Omega_{X'/Y'} = 0$ which follows from (5) and the corresponding statement for X/Y.

Proof of (10). If $f : X \to Y$ is an immersion, then it factors as $X \to V \to Y$ where $V \to Y$ is an open subspace and $X \to V$ is a closed immersion, see Morphisms of Spaces, Remark spaces-morphisms-remark-immersion. Let $V' \subset Y'$ be the open subspace whose underlying topological space $|V'|$ is the same as $|V| \subset |Y| = |Y'|$. Then $X' \to Y'$ factors through V' and we conclude that $X' \to V'$ is a closed immersion by part (1). This finishes the proof. $\square$